CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 11: The World of Algorithms

Every formula and result of the chapter on one sheet

1	Algorithm	A precise step-by-step procedure for solving a problem
2	Executing an algorithm	Carrying out its steps one by one
3	List	An ordered collection of values, written in square brackets
4	gcd (hcf)	The largest number that divides both numbers
5	Subtraction rule	$\gcd(m, n) = \gcd(n, m - n)$ for $m \geq n$
6	Remainder	$m \bmod n$ is the remainder when m is divided by n
7	Division rule	$\gcd(m, n) = \gcd(n, m \bmod n)$
8	Stopping rule	$\gcd(m, 0) = m$
9	lcm from the gcd	$\operatorname{lcm}(m, n) = \dfrac{m \times n}{\gcd(m, n)}$
10	Prime test	n is prime when divisors(n) has exactly two elements
11	Largest possible common divisor	$\min(m, n)$